

THE HERITAGE SCHOOL
FIRST TERM EXAMINATIONS, 2024-2025
SUBJECT –CHEMISTRY (PAPER 1)
CLASS -XII

FULL MARKS=70

TIME -3 hrs

*(Candidates are allowed **additional 15 minutes** for **only** reading the paper.*

*They must **NOT** start writing during this time).*

*This paper is divided into **four** sections – A, B, C and D.*

*Answer **all** questions.*

***Section A** consists of **one** question **having sub-parts** of **one** mark each. **Section B** consists of **ten** questions of **two** marks each.*

***Section C** consists of **seven** questions of **three** marks each, and **Section D** consists of **three** questions of **five** marks each.*

Internal choices have been provided in one question each in Section B, Section C and Section D.

All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.

The intended marks for questions or parts of questions are given in brackets []. Balanced equations must be given wherever possible and diagrams where they are helpful.

When solving numerical problems, all essential working must be shown. In working out problems, use the following data:

$$\begin{aligned}\text{Gas constant } R &= 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 8.314 \text{ JK}^{-1} \text{ mol}^{-1} \\ &= 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}\end{aligned}$$

$$1 \text{ l atm} = 1 \text{ dm}^3 \text{ atm} = 101.3 \text{ J. } 1 \text{ Faraday} = 96500 \text{ coulombs. Avogadro's number} = 6.023 \times 10^{23}.$$

SECTION A – 14 MARKS

Question 1

(A) Fill in the blanks by choosing the appropriate word(s) from those given in the bracket:

[less than, tetrahedral, low, octahedral, sp^3d^2 , sp^3 , sp^3d , dsp^2 , d^2sp^3 , formaldehyde, greater than, square planar, equal to, trigonal planar, high, ammonia, acetaldehyde, acetone, hydroxylamine, 2,4-DNP] **[4x1=4]**

- i. van't Hoff factor for sucrose is -----one while that for potassium ferrocyanide is -----one.
- ii. The geometry of $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ are -----and -----respectively.
- iii. Urotropine is formed by the action of ----- with -----
- iv. Outer orbital complexes involve -----hybridisation and are -----spin complexes.

(B) Select the correct alternative from the choices given below:

[7x1=7]

(i) Cyanohydrin of which compound gives lactic acid in hydrolysis?

- (a) Acetaldehyde (b) Formaldehyde (c) Acetone (d) Propanal

(ii) Phenol dimerises in benzene having van't Hoff factor 0.54. What is the degree of association?

- (a) 0.46 (b) 0.54 (c) 0.27 (d) 0.92

(iii) The reactivity order of 1° alkyl halides follows the order:

- (a) $\text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{X} < \text{CH}_3\text{CH}_2\text{CH}_2\text{X}$ (b) $\text{CH}_3\text{X} < \text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{CH}_2\text{CH}_2\text{X}$
(c) $\text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{CH}_2\text{CH}_2\text{X} < \text{CH}_3\text{X}$ (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{X} < \text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{X}$

(iv) Rapid conversion of α -D-glucose and β -D-glucose in solution is known as:

- (a) racemization (b) asymmetric induction (c) fluxional induction (d) mutarotation

(v) Which of the following will not show geometrical isomerism?

- (a) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (b) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (c) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ (d) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

(vi) Assertion: AgCN reacts with ethyl bromide to give ethyl cyanide.

Reason: CN^{-1} is an ambident nucleophile.

- (a) Both assertion and reason are true and the reason is the correct explanation of assertion.
(b) Both assertion and reason are true but reason is not the correct explanation for assertion.
(c) Assertion is true but reason is false.
(d) Assertion is false but reason is true.

(vii) Assertion: The complex $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$ does not give precipitate with AgNO_3 .

Reason: The given complex contains neutral ligands only.

- (a) Both assertion and reason are true and the reason is the correct explanation of assertion.
(b) Both assertion and reason are true but reason is not the correct explanation for assertion.
(c) Assertion is true but reason is false.
(d) Assertion is false but reason is true.

(C) Read the passage given below carefully and answer the questions that follow: [3]

The kinetic studies help us to know the speed or rate by which chemical reactions occur. For a chemical reaction:



The rate law has been found to be:

$$\text{Rate} = k[A][C]^2 \text{ with } k = 2.0 \times 10^{-5} \text{ mol}^{-2} \text{L}^2 \text{s}^{-1}$$

The initial concentrations of the reactants are $[A] = 0.1 \text{ mol L}^{-1}$, $[B] = 0.2 \text{ mol L}^{-1}$,
 $[C] = 0.4 \text{ mol L}^{-1}$

(i) What is the order of reaction?

(ii) How will the rate be affected when the concentration of all reactants A, B and C be doubled?

(iii) a. How will the rate of reaction be affected when the concentration of B is only doubled?

b. How will the rate be affected when the concentration of C alone is increased to
 1.2 mol L^{-1} ?

(iv) Calculate the initial rate of reaction after 0.04 mol L^{-1} of A has been reacted.

SECTION B – 20 MARKS

Question 2

[2]

i. K (the rate constant), E_a (Activation energy), and “A” (Arrhenius constant) are $3 \times 10^{-4} \text{ s}^{-1}$, $104.4 \text{ kJ mol}^{-1}$ and $6.0 \times 10^{14} \text{ s}^{-1}$ respectively. What is the value of “K” when $T \rightarrow \infty$?

ii. What type of molecules undergo effective collisions?

Question 3

[2]

An alcohol (A) with molecular formula $\text{C}_4\text{H}_{10}\text{O}$ gives instant turbidity with Lucas test. It reacts with P and I_2 to give an alkyl iodide (B) and with Na metal it gives sodium salt (C). When (B) and (C) are heated together, a new compound (D) with molecular formula C_4H_8 along with (A) is formed. Identify A to D with chemical equations.

Question 4

[2]

i. pK_a value of 4-Nitrobenzoic acid is lower than that of benzoic acid. Explain.

ii. Dialkyl cadmium is considered superior to Grignard reagent for the preparation of ketone from an acid chloride. Explain.

Question 5**[2]**

A first order reaction is 50% completed in 30 minutes at 300 K and in 10 minutes at 320 K. Calculate the activation energy of the reaction.

Question 6**[2]**

How can the following conversions be brought about?

i. Ethanal to propan-2-ol

ii. Phenol to benzoic acid.

Question 7**[2]**

i. FeSO_4 solution mixed with $(\text{NH}_4)_2\text{SO}_4$ solution in 1:1 molar ratio gives the test of Fe^{+2} but CuSO_4 solution when mixed with aqueous ammonia in 1:4 molar ratio does not give the test of Cu^{+2} ion. Explain why?

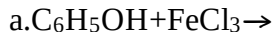
ii. Arrange the following complexes on the basis of their increasing order of conductivity in the solution : $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{Co}(\text{en})_3]\text{Cl}_3$, $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

Question 8**[2]**

An organic compound (A) has characteristic odour on treatment with NaOH it forms two compounds (B) and (C). Compound (B) has molecular formula $\text{C}_7\text{H}_8\text{O}$ which on partial oxidation gives back (A). The compound (C) is a sodium salt of an acid. When (C) is treated with soda lime it yields an aromatic hydrocarbon (D). Deduce the structures of (A), (B), (C), and (D).

Question 9**[2]**

(i) Write the main product formed in each case:



(ii) Arrange the following in increasing order of boiling point:



OR

(i) Write the chemical equation of the reaction of HI, at 373K with:

a. Methoxybenzene

b. 1-Propoxypropane

b. Arrange the following in the decreasing order of acid strength:

Ethanol, Isopropyl alcohol, Tertiary butyl alcohol

Question 10**[2]**

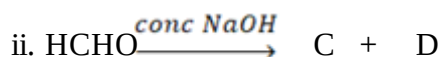
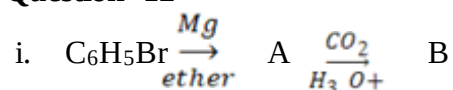
In a reaction $m\text{A} + n\text{B} \rightarrow \text{Products}$, when the concentration of A is doubled, the rate of reaction is also doubled. When the concentration of B is doubled, the rate of reaction becomes four times. Find the overall order of the reaction.

Question 11**[2]**

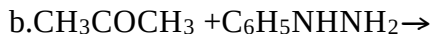
Write the chemical test to distinguish between the following pairs of compounds:

(i) Benzoic acid and phenol

(ii) Formic acid and acetic acid.

SECTION C-21 MARKS**Question 12****[3]****OR**

i. Give balanced chemical equations for the following:

**Question 13****[3]**

A solution containing 0.5 g of KCl dissolves in 100 g of water and freezes at -0.24°C . Calculate the degree of dissociation of the salt [k_f for water is $1.86 \text{ K Kg mol}^{-1}$]

Question 14**[3]**

Give balanced chemical equations with conditions where necessary for the following name reactions:

i. Rosenmund's reduction.

ii. Perkin reaction.

iii. Benzoin condensation.

Question 15**[3]**

i. $[\text{Fe}(\text{CN})_6]^{-4}$ is a coordination complex ion.

a. What is the oxidation number of iron in the complex?

b. How many unpaired electrons are there in the complex?

c. State the hybridisation and magnetic behaviour of the complex.

ii. On the basis of CFT, write the electronic configuration of d^4 ion if:

- a. $\Delta_0 > P$
 b. $\Delta_0 < P$

Question 16

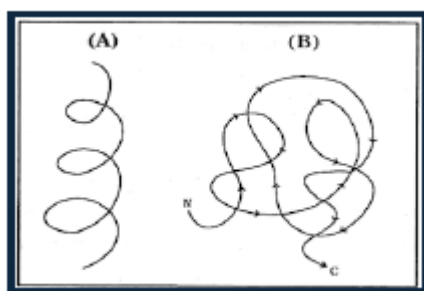
[3]

- i. Henry's law constant for methane in benzene at 298K is 4.27×10^5 mm of Hg. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg.
 ii. A solution of lactose containing 8.45 g of lactose in 100 g of water has a vapour pressure of 4.559 mm of Hg at 0°C . If the vapour pressure of pure water is 4.579 mm of Hg, calculate the molecular weight of lactose.

Question 17

[3]

- i. Give a reason for each given below:
 (a) Except glycine, all other naturally occurring α -amino acids are optically active.
 (b) Zwitter ion is also known as a dipolar ion.
 ii. Observe the structures of protein given below and answer the questions that follows:



- a. Which type of structures do **A** and **B** represent?
 b. How can these structures be converted to primary structure?

Question 18

[3]

Consider the data given below for the reaction $A + B \longrightarrow \text{Product}$

S.No.	conc. of [A] mol L ⁻¹	conc. of [B] mol L ⁻¹	Rate; mol L ⁻¹ sec ⁻¹
1.	1.0	0.15	4.20×10^{-6}
2.	2.0	0.15	8.40×10^{-6}
3.	1.0	0.20	5.60×10^{-6}

- i. What is the order of reaction with respect to A and B respectively?
 ii. State the rate law
 iii. Calculate the rate constant.

SECTION D-15 MARKS

Question 19

[5]

Alcohol and phenols are hydroxy derivatives of hydrocarbons. Both have the tendency to form intermolecular hydrogen bonding. Thus, they have higher boiling point and high solubility in water. As the O-H bond is polar in nature they also exhibit acidic property. Acidic characteristic depends upon inductive effect and resonance stabilization. The position and nature of substituents also affects the acidic strength.

(i) Select from each case as per the property mentioned in the bracket.

(a) o-nitrophenol, p-nitrophenol, m-nitrophenol (most acidic)

(b) o-cresol and m-cresol (least acidic)

(ii) Correct the given statement and give a reason for your answer.

The solubility of phenol is higher than alcohol.

(iii) Compounds [X] and [Y] are functional isomers of each other with molecular formula C_3H_8O .

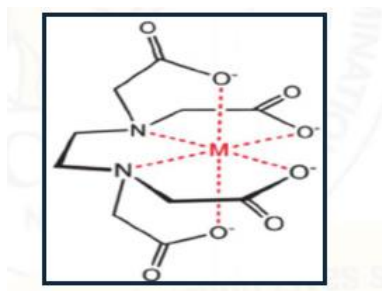
Draw the isomers.

(iv) Give balanced chemical equation for Kolbe reaction.

Question 20

[5]

i. Food and drug administration (FDA) has approved that Ethylenediamine tetra acetate (EDTA) is a medication in the treatment of lead poisoning. It is also used in the estimation of Ca^{2+} and Mg^{2+} during removal of hardness in water



i. Which type of a ligand is EDTA?

ii. Why EDTA is used in the treatment of lead poisoning?

iii. What is the geometry of EDTA complex?

iv. The complex $CoBr_3 \cdot 4NH_3 \cdot 2H_2O$ has molar conductivity corresponding to 3:1 electrolyte. Write the molecular formula and its IUPAC name.

v. Draw the geometrical isomers of $[Co(NH_3)_4Cl_2]^+$

OR

i. For the complex $[Ni(CO)_4]$, comment on-

(a) Oxidation state of Ni

(b) magnetic property of the complex

(c) hybridisation

(d) shape

ii. Give the IUPAC name of the following:

a. $[PtCl_2(NH_3)_4]$ $[PtCl_4]$

b. $[CoCl(ONO)(en)_2]^+$

iii. State the type of isomerism exhibited by the following complexes:

a. $[Co(NH_3)_5Cl]SO_4$

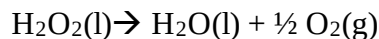
b. $[Co(en)_3]^{+3}$

iv. How does $K_2[PtCl_4]$ get ionised when dissolved in water? Will it form precipitate when $AgNO_3$ solution is added to it? Why?

Question 21

[5]

i. Aman and Rohan were working on the kinetics involved in decomposition of hydrogen peroxide in the presence of iodide ions.

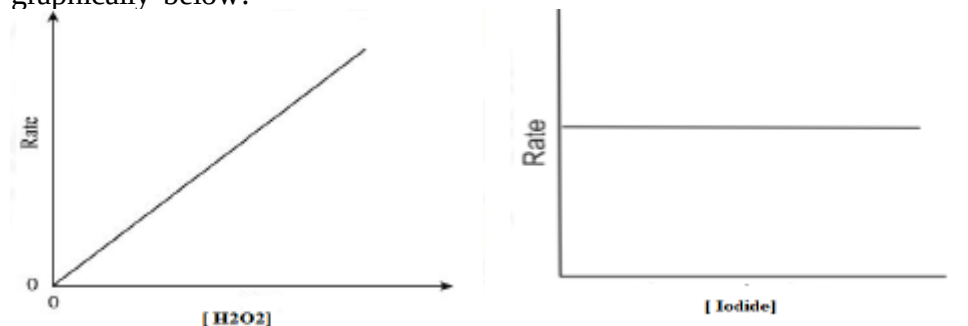


They both had their respective hypothesis as follows:

According to Aman the rate will be affected by changing the concentrations of hydrogen peroxide and iodide ions as iodide ions are involved in the reaction.

According to Rohan the rate will only be affected by changing the concentration of hydrogen peroxide.

They carried out several experiments to check their respective hypothesis. Their results are shown graphically below:



(i) What do you think whose hypothesis is correct and why?

(ii) Do you think that the half-life of this reaction will change if concentration of hydrogen peroxide is reduced to half. Give a reason.

(iii) In a chemical reaction, two reactants take part. The rate of reaction is directly proportional to the concentration of one of them and inversely proportional to the concentration of other. State the order of the reaction. What will be the unit of its rate constant if the concentration terms are in mol/L?

(iv) A 40% first order reaction is completed in 50 minutes. How much time will it take for the completion of 80% of the reaction?